Matt | Varghese Mathew

Software Systems Architect

Software Systems Architect specializing in large-scale distributed platforms, API ecosystems, and cross-system integrations. Proven track record of driving high-impact solutions across complex environments involving multiple teams, vendors, and evolving standards.

Ask Me || DM me on LinkedIn
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Madison, WI
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WORK EXPERIENCE

Software Systems Architect, Health Plans

Epic - Verona, WI — *MARCH 2021 - PRESENT*

Architect for large-scale systems enabling expedited decision workflows across hundreds of customers, reducing turnaround from days to near real-time.

- **Led architecture and delivery** of Epic's payer interoperability platform for CMS 2027, collaborating with internal teams, industry standards workgroups, and external vendors to refine and operationalize evolving standards.
- **Identified gap and established a unified "source of truth"** for prior authorization requirements, eliminating fragmentation across 12+ product modules and reducing new development lead time by ~80%.
- **Recognized need and architected standardized** data models and execution frameworks to replace inconsistent one-off implementations, significantly improving maintainability and cross-team scalability.
- **Modernized Medical Necessity guideline review integrations** using OAuth2 based single sign-on standards, enabling real-time clinical decision workflows and reducing integration defects by ~60%.
- **Mitigated tooling deficiencies by building simulation and testing frameworks** to emulate third-party systems, enabling end-to-end workflow validation, proactively eliminating defects, and accelerating partner onboarding.

Senior Software Systems Developer, Clinical Informatics

Epic - Verona, WI — *AUGUST 2010 - MARCH 2021*

Developed clinical decision support (CDS) integrations, web app integrations, and mobile push notifications. Directed adoption of modern healthcare interoperability standards and pioneered Epic's first high-availability cloud service on Azure.

- **Created Epic's high-availability mobile push notifications platform,** routing millions of weekly notifications via APNS and GCM. Pioneered deployment as one of Epic's first Azure-native production services, reducing operational costs by over 90%.

SKILLS

Architecture & Infrastructure

- SDLC
- Distributed Systems
- Microservices
- Azure
- CI/CD
- Docker
- Virtualization
- Hadoop
- Maintainable Architecture

Engineering Stack

- TypeScript
- React
- Web API
- Node
- .NET/C#
- C/C++
- Python
- Git & SVN
- ANSI-M
- SQL & NoSQL
- Linux & Windows

Specialized Expertise

- LLM API Integration
- Agentic Workflows
- Network Architecture & Routing

COLLABORATION

Standards-body Hackathons

Developed prototypes of industry specifications in collaboration with participants from other vendors to improve interoperability.

Prior Auth Burden Reduction

Represented Epic on public calls steering development of standards to reduce admin burden in Prior Authorization.

Authoring Industry Standards

Contributed to HL7 CDS Hooks and HL7 SMART Web Messaging specifications to enhance workflow integration and security.

- **Identified fragmentation and led enterprise-wide centralization** of 3rd-party web application integrations to establish standards for security and maintainability.
- **Designed & Developed Epic's decision support web services framework,** enabling standards-based clinical decision support across multiple clinical applications.
- **Identified usability gaps and redesigned** decision support configuration to align with mental models, improving usability and enabling efficacy feedback loops for doctors.
- **Recognized subject matter expert in Web App Security and RESTful integrations,** providing architectural guidance to multiple teams and onboarding external vendors.

KEY PROJECTS

AI Assistant for patients and doctors
2025 Q2

Integrates LLM APIs with electronic patient health record for agentic use-cases. LLM chat web app answers doctor and patient questions using real health data. Enables AI driven forms, AI decision support, AI care delivery, etc.

JSON Serializer & Deserializer for ANSI-M
2020

Spotlights long-term vision in creating powerful generic reusable tooling. Created RFC 8259 compliant JSON Serializer and Deserializer for ANSI-M. Serves as a foundation for 30+ database AI integrations across Epic.

PRIOR EXPERIENCE

Research Assistant, Condor Distributed Computing

University of Wisconsin - Madison, WI — *SEPT 2008 - MAY 2010*

Virtualized Condor jobs with VMware achieving deterministic runtime environments.

Software Engineer, Enterprise Routing

Cisco Systems - Bangalore, India — *JUNE 2006 - JULY 2008*

Platform engineering and configuration tooling for Cisco's enterprise class routers.

EDUCATION

Master of Science, Computer Science

University of Wisconsin - Madison — *2008-2010*

GPA: 3.94 / 4.0 | Focus: Systems, Networking, Distributed Computing.

Bachelor of Technology, Computer Science and Engineering

National Institute of Technology - Calicut, India — *2002-2006*

CGPA: 9.60 / 10.0 | Department Topper and Gold Medalist.

Epic Vendor Conferences

Presented for several years on Epic's web service integration capabilities to educate and onboard third-party vendors.

Epic Mentoring

Mentored engineers and interns across multiple teams, driving adoption of architectural best practices and development of high-performing teams.

Epic Recruitment

Attracted top talent to Epic by representing Epic at university career fairs and conducting candidate interviews.

GITHUB

Web Based SSH Client
2025 Q3

Non-traditional solution for real world cost reduction avoiding remote-app licenses for moderated shell access to servers.

Demonstrated proof of concept fully browser-based SSH client for unix servers.

github.com/mattvarghese/WebSSHClient

HTTP Auto-responder
2024 Q4

Development tooling for Web APIs, keeping traffic within the enterprise or local network.

Configurable endpoint allows simultaneous multi-user testing of RESTful APIs.

github.com/mattvarghese/HttpResponder

MIPS Pipeline Simulator
2005, 2025

Demonstrates deep expertise in low-level systems, architecture, parallelism, and performance.

5-Stage pipeline simulation of a MIPS microprocessor, with cache simulation, assembler, and terminal app developed with POSIX APIs, C++, Flex, and Bison in 2005.

In 2025, developed a Small-C compiler above the assembler with AI assistance.

github.com/mattvarghese/coconut

OTHER

- USHPA Novice (H2) rated hang-glider pilot
- Solo endorsed VFR student pilot
- ASA 101,103/104,105 rated sailor
- Learned Elementary Russian as an adult
- Avid reader, ice-skating, volleyball, XC-skiing, bicycling